Bank Marketing Machine Learning Engineering Report

End-to-End Rigorous Preprocessing, Stratified Split Protocol & Dual-Architecture Deployment

1. Executive Summary & Objective

This empirical report formalizes the complete data engineering, exploratory data analysis (EDA), and machine learning preprocessing pipeline for the **Bank Marketing Dataset** (*bank-additional-full.csv*). The dataset records direct marketing phone campaigns of a Portuguese banking institution targeting client subscriptions to long-term deposits (target variable *y*). The core objective of this architecture is to establish a rigorous, mathematically validated pipeline that guarantees **zero data leakage**, handles missing and irregular values via distance-weighted KNN imputation, rectifies severe parametric skewness through Yeo-Johnson power transformations, and addresses class imbalance via Synthetic Minority Over-sampling Technique (SMOTE). Crucially, the pipeline yields two distinct, production-grade training corpora: training_preprocessed.csv and training_smote.csv, each engineered for distinct classifier families.

2. Zero-Leakage Stratified Splitting Protocol

A foundational tenet of production machine learning is the absolute separation between training data and evaluation benchmarks. To enforce this, the raw corpus of 41,188 observations was partitioned using a strict **70% Training / 15% Test / 15% Validation** stratified protocol based on the binary target *y*. All subsequent scalers, imputers, power transformers, and categorical encoders were fitted **exclusively on the training split**. The test and validation partitions were saved immediately as raw splits and strictly quarantined, guaranteeing that out-of-sample generalization metrics will reflect true real-world predictive validity.

Partition File	Proportion	Total Rows	No ('y=0')	Yes ('y=1')	Conversion Rate
training.csv	70.0%	28,831	25,583	3,248	11.27%
test.csv	15.0%	6,178	5,482	696	11.27%
validation.csv	15.0%	6,179	5,483	696	11.26%
Total Raw Corpus	**100.0%**	**41,188**	**36,548**	**4,640**	**11.27%**

3. Step-by-Step Training Preprocessing Pipeline

Every transformation applied to training.csv addresses specific empirical characteristics of bank marketing data:

- **Step A — Target Standardization:** Target variable *y* encoded as binary integer (no: 0, yes: 1). Baseline positive class prevalence is 11.27%, confirming significant class imbalance.
- **Step B — Unknown & Irregular Coding:** Values coded as *'unknown'* in demographic and credit attributes (job, marital, education, default, housing, loan) converted to formal missingness (np.nan). In pdays, code 999 ('client not previously contacted') replaced with np.nan, and a binary indicator pdays_contacted (1 if contacted, 0 if 999) engineered to capture contact precedence.
- **Step C — Ordinal & Binary Harmonization:** education mapped to an integer ordinal continuum from 0 (illiterate) to 6 (university degree) based on human capital hierarchy. Binary variables (contact, housing, loan, default) encoded as standard 0/1 integers.
- **Step D — Domain Feature Engineering:** Engineered 7 high-signal clinical attributes: (1) campaign_capped (clipped at 10 to truncate extreme long-tail fatigue), (2) contact_frequency_ratio (*previous / (campaign + 1)*), (3) economic_pressure_index (standardized composite score from *emp.var.rate*, *euribor3m*, and *nr.employed*), (4) age_bucket (5 life-stage brackets), (5) duration_bucket (3 call engagement tiers), and (6) month_quarter (seasonal quarter).
- **Step E — Multivariate KNN Imputation:** Missing numeric and ordinal values (housing, loan, default, education_ordinal, pdays, age_bucket) imputed via distance-weighted KNN (*k=7*, Euclidean metric). Mode imputation utilized for remaining high-cardinality nominals.
- **Step F — One-Hot Categorical Expansion:** Categorical features (job, marital, month, day_of_week, poutcome) converted into clean binary indicator vectors with standardized prefixes, preventing false ordinal ordering in linear and neural architectures.
- **Step G — Yeo-Johnson Power Transformation:** Applied Yeo-Johnson transformation across all 12 continuous features to stabilize variance, compress positive skewness, and project continuous regressors onto quasi-Gaussian distributions.

4. Mathematical Foundations: Yeo-Johnson & SMOTE

Yeo-Johnson Transformation Formulation:
Unlike Box-Cox, Yeo-Johnson supports zero and negative values, crucial for macro indicators like *emp.var.rate*. The transformation function $\psi(\lambda, y)$ is defined piecewise:

$$\psi(\lambda, y) = \begin{cases} \frac{(y + 1)^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, y \geq 0 \\[4pt] \ln(y + 1) & \text{if } \lambda = 0, y \geq 0 \\[4pt] -\frac{(-y + 1)^{2 - \lambda} - 1}{2 - \lambda} & \text{if } \lambda \neq 2, y < 0 \\[4pt] -\ln(-y + 1) & \text{if } \lambda = 2, y < 0 \end{cases}$$

Parameter λ is estimated via profile log-likelihood maximization. Following transformation, features are standardized to zero mean and unit variance $(\mu=0, \sigma=1)$.

SMOTE (Synthetic Minority Over-sampling Technique) Formulation:
To balance minority class instances $(y=1)$ without duplicate overfitting, SMOTE interpolates synthetic instances in d-dimensional feature space. For each minority sample x_i, its k nearest neighbors in the minority class $\{x_{i(1)}, \dots, x_{i(k)}\}$ are computed via Euclidean metric. A neighbor x_{zi} is selected at random, and a synthetic sample x_{new} is generated along the line segment:

$$x_{new} = x_i + \delta \cdot (x_{zi} - x_i), \quad \text{where } \delta \sim \text{Uniform}(0, 1)$$

5. Architectural Comparison: training_preprocessed vs. training_smote

The pipeline creates two distinct training artifacts. Selecting the appropriate artifact is critical for maximizing downstream predictive performance:

Feature / Attribute	training_preprocessed.csv	training_smote.csv
Sample Count	28,831 observations	51,160 observations (25,580 Class 0 / 25,580 Class 1)
Class Distribution	Natural Imbalanced: 88.73% No / 11.27% Yes	Perfectly Balanced: 50.0% No / 50.0% Yes
Marginal Priors	Preserves true population distribution & base rates	Priors artificially altered; probability calibration shifted
Recommended Classifiers	Gradient Boosting (XGBoost, LightGBM, CatBoost), Random Forests, Cost-Sensitive Classifiers	Deep Neural Networks (MLP), Support Vector Machines (RBF Kernel), Logistic Regression, k-NN
Primary Advantage	Optimal Bayesian posterior probabilities and calibration	Mitigates gradient starvation for minority class in backprop
Optimal Loss / Metric	Focal Loss, scale_pos_weight, PR-AUC, Brier Score	Cross-Entropy Loss, ROC-AUC, Balanced Accuracy, F1-Score

6. Operational Deployment Directives

When to Deploy training_preprocessed.csv:
Deploy this file whenever utilizing tree-based ensemble algorithms (XGBoost, LightGBM, CatBoost). Tree partitions split along feature thresholds and are invariant to monotonic transformations, while parameterizing scale_pos_weight = 7.87 (the ratio of negative to positive cases) natively handles class imbalance without distorting feature manifolds. Additionally, financial risk scoring systems requiring accurate calibrated probabilities (e.g. expected value of marketing outreach) must train on this set to maintain unskewed base rate priors.

When to Deploy training_smote.csv:
Deploy this file when training gradient-descent algorithms (Multi-Layer Perceptrons, Deep Tabular Networks, Logistic Regression) or distance-based models (SVMs, k-Nearest Neighbors). In un-resampled imbalanced data, batch loss gradients are dominated by majority-class errors, causing stochastic gradient updates to collapse into trivial majority-class classifiers. SMOTE guarantees balanced gradient vectors across mini-batches, enforcing active decision boundary construction in sparse minority feature regions.

Critical Evaluation Mandate:
Regardless of whether a model was trained on training_preprocessed.csv or training_smote.csv, **all final model performance evaluation must be conducted strictly on the untouched test.csv partition**. Synthetic samples must never contaminate test or validation sets. Evaluate models using Precision-Recall Area Under Curve (PR-AUC), F1-Score at optimal threshold, and Cost-Weighted Confusion Matrices rather than raw classification accuracy.

7. Comprehensive Pipeline Output Manifest

Artifact Path	Type	Dimensions	Description
data3/training.csv	Raw Split	28,831 x 21	70% stratified raw training data (source for all fitting)
data3/test.csv	Raw Split	6,178 x 21	15% quarantined evaluation benchmark (strictly untouched)
data3/validation.csv	Raw Split	6,179 x 21	15% quarantined hyperparameter validation benchmark
data3/training_preprocessed.csv	Transformed	28,831 x 67	Engineered, KNN-imputed, Yeo-Johnson normalized training set
data3/training_smote.csv	Balanced	51,166 x 67	SMOTE 50:50 balanced dataset for neural networks & SVMs
data3/Plots/Raw/	Visualizations	15 High-DPI PNGs	Distributions, financial flags, and correlations of raw data
data3/Plots/Preprocessed/	Visualizations	20 High-DPI PNGs	Normalized distributions, OHE categories, and heatmaps
data3/Plots/SMOTE/	Visualizations	20 High-DPI PNGs	Balanced distributions, synthetic manifolds, and heatmaps
data3/new_report.pdf	Documentation	Formal PDF	Complete academic engineering report & deployment guide